

A physically-informed two-stage approach to day-ahead, 15-minute wind power forecasting for a research-instrumented small wind turbine

Case study: the Chalmers wind turbine, Björkö, Sweden

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Abstract

Day-ahead scheduling of small, grid-connected wind turbines requires power forecasts at fine temporal resolution, but many research turbines lack the multi-year, gap-free SCADA records that conventional forecasting pipelines assume. This paper presents a physically-informed, two-stage forecasting approach, in which numerical weather prediction (NWP) wind speed forecasts are converted to power through a turbine-specific empirical power curve, developed for the Chalmers wind turbine on the island of Björkö, Sweden, a 45 kW research platform instrumented far beyond typical commercial density. Using a publicly archived high-frequency SCADA dataset, we derive a monotonic empirical power curve from 20 Hz measurements restricted to confirmed normal-operation controller states, fit a linear bias correction between historical NWP wind speed and mast-measured wind speed, and validate the pipeline with leave-one-day-out cross-validation across 44 non-continuous measurement campaigns spanning July 2022 to August 2023. The bias-corrected pipeline achieves an out-of-sample normalized RMSE of 0.274 relative to the fitted plateau power, with wind-speed bias reduced from -0.96 m/s to near zero. We show that the dominant remaining error arises from the nonlinear amplification of wind-speed error through the steep ramp region of the power curve rather than from the wind forecast itself, and we discuss the structural limitations imposed by non-continuous data collection and by the use of a historical forecast proxy rather than true archived day-ahead forecast runs.

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1. Introduction

Accurate day-ahead forecasts of wind power output are a prerequisite for grid balancing, market participation, and operational planning at any scale of wind generation. For utility-scale wind farms, forecasting pipelines typically draw on years of continuous SCADA logging paired with archived numerical weather prediction (NWP) output. Research and demonstration turbines, however, are frequently operated in discrete measurement campaigns rather than continuously, and their available historical weather data may likewise be limited to reconstructed or short-lead-time forecast products rather than true archived operational forecasts. This paper documents the design, implementation, and validation of a day-ahead, 15-minute-resolution power forecasting pipeline built under exactly these constraints, using the Chalmers wind turbine at Björkö as a case study.

We adopt a two-stage forecasting strategy: wind speed and direction are forecast independently using operational NWP products, and the resulting wind speed is converted to expected electrical power through an empirically fitted, turbine-specific power curve. This design is chosen deliberately over a direct, end-to-end statistical mapping from weather variables to power, for three reasons relevant to this dataset. First, it is modular: the meteorological forecast and the turbine response model can be validated, replaced, or improved independently. Second, it is interpretable: any forecast error can be attributed to either the weather forecast or the turbine conversion step rather than absorbed into an opaque model. Third, and most importantly for this case, it is tractable given the available historical record: the turbine's own measurement history consists of 44 non-continuous days over a thirteen-month span, which is insufficient to train a sequence model capable of learning genuine temporal dynamics, but is sufficient to characterize the turbine's static wind-to-power response with high statistical confidence.

The remainder of this paper is organized as follows. Section 2 describes the turbine and site. Section 3 describes the available SCADA dataset and the data-quality considerations that shaped our approach. Section 4 details the forecasting methodology: the empirical power curve, the NWP wind speed sources used for live and historical forecasting, the temporal disaggregation from hourly to 15-minute resolution, and the bias-correction step. Section 5

presents results, including an honest, cross-validated estimate of forecast skill. Section 6 discusses the sources of residual error and the structural limitations of the study, and Section 7 concludes.

2. Site and turbine description

The Chalmers wind turbine is a 45 kW research and demonstration turbine situated on the island of Björkö, in Gothenburg's northern archipelago, at coordinates 57.71819°N, 11.68338°E, approximately 20 km west of central Gothenburg. The turbine has been in its current configuration since 2021 and is designed, in its entirety, by Swedish engineers, with the exception of a generator sourced from the United Kingdom under a previous EU project. It is grid-connected and produces electricity whenever it is operating.

Distinctive structural features of the turbine, relevant to interpreting its measured behavior, include a 30 m tower constructed of wood (a modular prototype intended to reduce embodied CO₂ relative to conventional steel towers), a concrete foundation with instrumented iron reinforcement, and fully carbon-fiber blades chosen to minimize rotor mass. The rotor diameter is 15.9 m, with 7.5 m blades under individual electric pitch control. The turbine operates at variable speed via a direct-driven generator and full-scale frequency converter, with rated speed of 75 rpm and rated power of 45 kW.

The turbine's measurement and control system is built around National Instruments CompactRIO hardware programmed in LabVIEW, with an EtherCAT bus connecting the turbine controller to expansion chassis in the nacelle and tower. The control loop samples at a 10 ms period, with a subset of channels sampled at 5 ms, giving the turbine substantially higher sensor density than is typical for a commercial unit of this scale. This instrumentation density is what makes the turbine a useful research asset, and it is also the practical reason the archived dataset used in this study is organized as discrete high-frequency measurement campaigns rather than continuous low-rate SCADA logging (Section 3).

3. Data

3.1 Source dataset

The turbine's measurement history used in this study is drawn from a publicly archived dataset, “Björkö Wind Turbine Version 1 (45kW) high frequency Structural Health Monitoring (SHM) data” (Fogelström et al., 2023), hosted on Zenodo under a Creative Commons Attribution 4.0 license (DOI:

10.5281/zenodo.8230330). The archive contains 69 SCADA and structural channels — including nacelle accelerations and tower/blade bending moments — sampled at two rates: 100 Hz and 20 Hz.

Initial inspection showed that the 100 Hz product (file B1_CL4_100.csv) consists of only nine short structural-health-monitoring bursts recorded on four distinct days, totalling roughly 330,000 rows, and is not representative of continuous turbine operation. The 20 Hz product (file B1_CL4_20.csv, approximately 2.8 GB) is the usable operational record: it spans 5 July 2022 to 2 August 2023 but is likewise organized as discrete measurement campaigns rather than continuous logging, covering 44 distinct calendar dates.

3.2 Channel selection and operating-mode filtering

Timestamps in the raw archive are recorded as LabVIEW time (seconds since 1904-01-01 UTC). Wind speed is available from two anemometers: WS30, mounted on a 30 m meteorological mast, and WSN, a nacelle-mounted anemometer whose readings are frequently near zero while the turbine is producing power, consistent with it sitting in the rotor's induction/wake zone. WS30 was therefore used as the wind speed reference throughout this study. There is no dedicated power channel in the archive; instantaneous electrical power is reconstructed as the product of rectifier current and rectifier voltage ($DCC \times DCV$), giving DC-side power at the rectifier rather than AC power exported to the grid. A separate channel, MaxPwrEst, is a fast-frequency-response power estimate rather than a measurement, and was not used; similarly, a WindEst channel was found to be effectively unused, frequently stuck at a fixed value.

The controller's operating state is recorded in a SysMode enumeration, mapped via the archive's accompanying metadata. Normal power production corresponds to SysMode = 12 (“Running”); all other codes correspond to initialization, stop, start, service, park, or emergency states in which measured power is at or near zero regardless of wind speed. All power-curve fitting and forecast validation in this study is restricted to SysMode = 12 samples, to avoid contaminating the turbine's wind-to-power response with non-operational periods.

3.3 Data-quality implications for methodology

Two properties of this dataset directly shaped the forecasting approach adopted in Section 4. First, restricting to Running-mode 20 Hz samples yields a very large and well-populated sample for characterizing the turbine's static

power curve (hundreds of thousands of samples in the 4–10 m/s range), making the empirical curve statistically robust despite the campaign-style collection. Second, aggregating Running-mode samples into completed 15-minute windows yields only 311 quarter-hour observations spread across 44 non-contiguous days, with gaps of up to several weeks between campaigns. This second property rules out any modeling approach that depends on learning continuous temporal dynamics (autocorrelation across an unbroken time series, diurnal cycles, multi-day persistence), and motivated the two-stage, physically-informed design described next.

4. Methodology

4.1 Empirical power curve

An empirical power curve was constructed by binning Running-mode WS30 samples into 0.5 m/s wind-speed bins and computing the mean measured power ($DCC \times DCV$) within each bin. Bins with fewer than 2,000 samples were treated as statistically unreliable and excluded from curve fitting, which removed both the lowest wind-speed bins (where residual DC power reflects rectifier standby behaviour rather than a physical aerodynamic response) and the sparsest high-wind-speed bins.

The cut-in wind speed was identified programmatically by locating the local minimum (“trough”) immediately preceding the monotonic rise in bin-mean power, rather than assuming a fixed manufacturer value. A monotonic piecewise cubic Hermite interpolating polynomial (PCHIP) was then fit to the well-populated bins, after forcing the bin means to be non-decreasing with wind speed (via a running maximum) to remove small non-physical dips caused by sampling noise in sparser bins. The resulting function,

$P(v) = PCHIP\text{-fit}(v)$, clipped to 0 below cut-in and to a fitted plateau value above the highest well-populated bin,

was saved as a reusable callable, together with its defining knots, for use in the forecasting pipeline.

4.2 NWP wind speed forecasts

Two independent NWP data sources were used, corresponding to two distinct use cases. For operational, forward-looking day-ahead forecasting, wind speed and direction are pulled from the Swedish Meteorological and Hydrological Institute's (SMHI) open forecast API, which provides an approximately ten-day-ahead forecast at the queried point, updated continuously. The originally

targeted SMHI product (PMP3g version 2) was retired on 31 March 2026; the pipeline was adapted to fail over to its replacement, SNOW1gv1, with parameter names remapped accordingly.

For retrospective validation against the turbine's 2022–2023 measurement campaigns, SMHI's live forecast cannot be used, since it does not retain historical forecast runs. Historical wind speed was instead obtained from Open-Meteo's Historical Forecast API, which reconstructs a continuous hourly time series by stitching together the earliest forecast hours of successive operational model runs. This product is a reasonable proxy for realistic forecast behaviour — unlike reanalysis products, it is built from actual forecast model output rather than being adjusted with hindsight knowledge of observed conditions — but it is not equivalent to retrieving the exact 24-hour-ahead forecast that would have been issued on a specific past day. True single-model-run archives with full forecast-horizon reproducibility (e.g., Open-Meteo's Single Runs API) are only available from 2024 onward and therefore do not cover the turbine's 2022–2023 measurement campaigns. This limitation is treated explicitly as a caveat throughout the validation results in Section 5.

4.3 Temporal disaggregation

Both NWP sources natively provide hourly wind speed values, while the forecasting target resolution is 15 minutes. Disaggregation is performed by linear interpolation in time between hourly values, which preserves the exact hourly forecast at the top of each hour and interpolates the three intermediate quarter-hour steps.

An optional variability-injection layer was also developed, motivated by the observation that linear interpolation understates real intra-hour wind speed variability. Two variability statistics were extracted from the Running-mode 20 Hz record: the standard deviation of wind speed within completed 15-minute windows (median 0.83 m/s), and the standard deviation of 15-minute mean wind speeds around their enclosing hourly mean, computed only for hours with at least three quarter-hour observations (0.41 m/s). The former characterizes sub-15-minute turbulence and is too large in magnitude to add to already-averaged 15-minute values; the latter characterizes exactly the inter-quarter-hour scatter relevant to hourly-to-15-minute disaggregation and was used as the default noise amplitude. When enabled, mean-zero noise scaled by this profile (as a function of local wind speed) is added to the three interior

quarter-hour points of each hour, while the values at the top of each hour are left exactly equal to the NWP forecast, so that the disaggregation neither drifts from nor biases the underlying hourly forecast.

4.4 Bias correction (model output statistics)

Comparison of the historical NWP wind speed against observed WS30 across all 311 aligned Running-mode quarter-hours revealed a systematic underestimation: the historical forecast underestimates mast wind speed by 0.957 m/s on average, a bias that is not visible in magnitude-only error metrics such as mean absolute error. An ordinary least squares regression was fit across the aligned points,

$$\text{observed WS30} = 0.930 \times \text{forecast wind speed} + 1.356 \quad (R^2 = 0.65),$$

and applied as a linear correction to forecast wind speed, clipped at zero, before conversion through the power curve. This is a standard model output statistics (MOS) approach: rather than attempting to correct the underlying NWP model, a simple site-specific statistical adjustment is learned from the available paired observations.

4.5 Validation protocol

Because the bias-correction coefficients are themselves fit from the same 311 observations used to evaluate the pipeline, an in-sample evaluation of the correction would overstate its expected benefit on unseen data. To obtain an honest, out-of-sample estimate of forecast skill, we used leave-one-day-out (LODO) cross-validation across the 44 measurement-campaign days: for each day, the bias-correction coefficients were refit using only the remaining 43 days, and the refit correction was applied to the held-out day before scoring. All 311 held-out predictions across the 44 folds were then pooled to compute aggregate error metrics. The coefficients ultimately shipped in the operational pipeline are those fit on all 44 days, with the LODO result serving as the honest estimate of how that fit is expected to generalize to a new, unseen day.

Forecast skill is reported using mean absolute error (MAE), root mean squared error (RMSE), and normalized RMSE (nRMSE), with normalization taken against the fitted plateau power (19.99 kW) rather than the turbine's 45 kW nameplate rating, since the turbine's measured output in this dataset does not approach nameplate capacity (Section 6).

5. Results

5.1 Empirical power curve

The fitted power curve exhibits cut-in at 3.0 m/s and reaches a plateau of 19.99 kW above 13.75 m/s. Table 1 summarizes the well-populated bins used in the fit.

Wind bin (m/s)	Mean WS30 (m/s)	Mean power (kW)	Samples
3.25	3.27	3.47	166,000
4.25	4.24	4.79	302,000
5.25	5.25	7.98	297,000
6.25	6.25	11.67	375,000
7.25	7.24	15.31	304,000
8.25	8.24	18.45	231,000
9.25	9.24	19.97	197,000
10.25	10.23	19.40	115,000
12.25	12.22	18.59	22,000

Table 1. Empirical power-curve bins (20 Hz, Running mode, 0.5 m/s bins) used to fit the monotonic PCHIP curve.

The measured plateau of approximately 20 kW is well below the turbine's 45 kW nameplate rating. This gap is discussed in Section 6; it is treated throughout this study as the turbine's genuine, real-world operating envelope during the measurement campaigns rather than as a data or fitting error.

[Insert Figure 1 here: power_curve_pchip_fit.png — fitted PCHIP curve overlaid on raw bin means.]

5.2 Baseline (uncorrected) forecast performance

Across all 44 measurement-campaign days, 311 fifteen-minute Running-mode observations were successfully aligned with historical NWP-derived forecasts. Table 2 reports aggregate error both for wind speed (historical NWP forecast vs. observed WS30) and for power (forecast wind speed through the fitted curve vs. observed power), prior to any bias correction.

Metric	Wind speed	Power
MAE	1.289 m/s	4.599 kW
RMSE	1.571 m/s	5.776 kW
nRMSE	0.235 (vs. 6.68 m/s mean obs.)	0.289 (vs. 19.99 kW plateau)

Table 2. Uncorrected pipeline performance, 311 aligned points across 44 campaign days.

Wind-speed error is moderate in absolute terms, but the power-curve mapping amplifies it substantially in the 4–9 m/s range — the region where the curve is steepest and where the majority of campaign observations fall — so a wind speed miss of roughly 1.3 m/s typically becomes several kilowatts of power error.

5.3 Bias-corrected performance

Applying the linear bias correction (Section 4.4) removed the systematic wind-speed underestimation entirely and produced a modest but consistent improvement in power accuracy, both in-sample and under honest leave-one-day-out cross-validation (Table 3).

Metric	Uncorrected	In-sample corrected	LODO out-of-sample
Wind MAE (m/s)	1.289	0.949	1.003
Wind RMSE (m/s)	1.571	1.239	1.315
Wind bias (m/s)	−0.957	0.000	+0.009
Power MAE (kW)	4.599	4.224	4.286
Power RMSE (kW)	5.776	5.454	5.479
Power nRMSE	0.289	0.273	0.274
Power bias (kW)	−1.931	+1.076	+1.084

Table 3. Three-way comparison of uncorrected, in-sample corrected, and out-of-sample (LODO) pipeline performance.

The out-of-sample result is only marginally worse than the in-sample result, indicating the correction generalizes well and is not an artifact of fitting to the evaluation data. Notably, the sign of the power bias flips from negative (under-prediction) before correction to positive (over-prediction) after correction. This is a consequence of the power curve's convexity in the ramp region rather than a modeling error: because the curve maps a wind-speed input nonlinearly to a power output, removing the mean wind-speed bias does not remove the mean power bias, and shifting the wind-speed distribution upward moves more mass into the steep part of the curve, producing a small net over-prediction in power.

5.4 Cross-validation stability

Table 4 summarizes the stability of the fitted bias-correction coefficients across the 44 leave-one-day-out folds. The slope coefficient a is highly stable (approximately 2% relative standard deviation), while the intercept b varies somewhat more in absolute terms, though within a range of roughly 0.8 m/s. This stability, combined with the near-zero out-of-sample wind bias, supports using the coefficients fit on all 44 available days as the coefficients shipped in the operational pipeline.

Coefficient	Mean	Std	Min	Max
a (slope)	0.930	0.020	0.884	1.037
b (intercept, m/s)	1.356	0.104	0.850	1.682

Table 4. Fold-to-fold stability of bias-correction coefficients across 44 leave-one-day-out folds.

[Insert Figure 2 here: backtest_power_scatter.png — forecast vs. observed power, all aligned points.]

[Insert Figure 3 here: backtest_power_timeseries.png — observed vs. forecast power overlay for representative campaign days.]

6. Discussion

6.1 Decomposing the residual error

Separating wind-speed error from power-curve error (Table 2 and Table 3) shows that the majority of remaining power forecast error is not attributable to the bias correction leaving residual wind-speed error uncorrected, but to the intrinsically nonlinear amplification of any wind-speed error through the steep ramp region of the power curve. Improving the linear wind correction further (e.g., with a nonlinear or wind-direction-dependent correction) is therefore likely to yield diminishing returns; the larger opportunity for improvement lies in narrowing the gap between the empirical power curve and the turbine's true instantaneous response, for example by incorporating turbulence intensity or wind direction as additional predictors of power at a given mean wind speed.

6.2 Why the plateau is well below nameplate rating

The turbine's fitted plateau of approximately 20 kW, against a 45 kW nameplate rating, likely reflects several compounding factors specific to a research turbine of this kind: possible yaw misalignment during some campaigns, the 30 m met-mast measurement height not exactly matching hub height, deliberate curtailment or fast-frequency-response (FFR) power

limitation associated with the turbine's grid-services research program, and the use of DC rectifier power rather than AC power actually exported to the grid, which excludes inverter losses in the opposite direction (i.e., rectifier power somewhat overstates true grid export, partially offsetting the other factors). None of these factors invalidate the forecasting approach; rather, they are the reason an empirical, data-derived power curve was used in preference to a theoretical or manufacturer-specified curve, since the empirical curve captures the turbine's actual delivered behaviour rather than an idealized one.

6.3 Structural limitations

- Non-continuous measurement record: the validation set consists of 311 quarter-hour observations across 44 non-contiguous days spanning thirteen months, not a continuous year of operation. Confidence intervals on the reported skill metrics should be interpreted accordingly.
- Historical forecast proxy: retrospective validation uses a stitched short-lead-time historical forecast product rather than a true archived day-ahead single model run, since single-run archives with full forecast-horizon reproducibility are only available from 2024 onward. Reported skill should be read as indicative of pipeline behaviour rather than a certified day-ahead forecast score.
- Uneven sample density across the wind-speed range: several of the highest wind-speed bins used in the power-curve fit, and several wind-speed ranges in the sub-hourly variability profile, are supported by relatively few independent samples and should be treated with additional caution.
- Single-site, single-turbine result: all findings are specific to this turbine, this site, and this measurement campaign, and should not be assumed to generalize to other turbines or coastal sites without independent recalibration.

7. Conclusion and future work

This study developed and validated a day-ahead, 15-minute-resolution wind power forecasting pipeline for a research-instrumented small wind turbine, under the constraint of a non-continuous historical measurement record. A two-stage, physically-informed design — independent NWP wind speed forecasting converted through an empirically fitted power curve — was chosen specifically because it remains tractable when continuous SCADA history is unavailable, in contrast to end-to-end statistical approaches that require it. A

simple linear bias correction, validated by leave-one-day-out cross-validation, meaningfully reduced wind-speed forecast bias and produced a modest, genuinely out-of-sample improvement in power forecast accuracy, reaching a normalized RMSE of 0.274 relative to the turbine's fitted plateau output.

Natural extensions of this work include probabilistic forecasting — using the residual distribution characterized during cross-validation to construct quantile bands around the point forecast rather than a single deterministic value — and incorporation of wind direction and turbulence intensity as additional predictors of the turbine's power response. As the turbine's measurement record continues to grow, particularly if future campaigns increase temporal continuity, revisiting a direct, data-driven power-forecasting model would become increasingly viable and could be compared directly against the physically-informed baseline established here.

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